

First understand the materials that may be used to create a development environment.

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Next, follow the tutorial below to correctly create the development environment.

1. Install the Arduino IDE

Arduino software

The Arduino Integrated Development Environment (IDE) is the software side of the Arduino platform. Used to write and upload code to the dashboard . Follow the tutorial to install the Arduino software (IDE) .

1.1 Enter the Arduino software official website

Enter in the browser and click to go to <https://www.arduino.cc/en/software> webpage, you can see the following webpage location:

[HARDWARE](#) [SOFTWARE](#) [CLOUD](#) [DOCUMENTATION ▼](#) [COMMUNITY ▼](#) [BLOG](#) [ABOUT](#)

Downloads



Arduino IDE 2.0.0

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger.

For more details, please refer to the [Arduino IDE 2.0 documentation](#).

DOWNLOAD OPTIONS

Windows Win 10 and newer, 64 bits
Windows MSI installer
Windows ZIP file

Linux AppImage 64 bits (X86-64)
Linux ZIP file 64 bits (X86-64)

macOS 10.14: "Mojave" or newer, 64 bits

(Here, take win10 system to install version 2.0.0 IDE as an example. For lower systems, please slide the webpage below to install version 1.8.X software. At the same time, when you see this tutorial, there may be a newer version on the website!)

1.2 Select the system version for software adaptation

Select the development software compatible with your computer system to download, here take Windows 10 as an example.



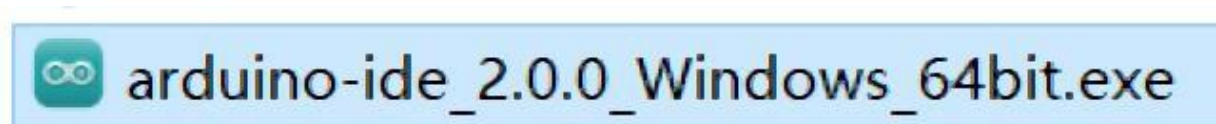
You can choose between an installer (.exe) and a Zip package. We recommend that you use the first "Windows Win10 and newer" to directly install everything you need to use the Arduino software (IDE), including drivers. Whereas with the Zip package, you need to install the drivers manually. Of course Zip files are also useful if you want to create a portable installation.

Click "Windows Win10 and newer"



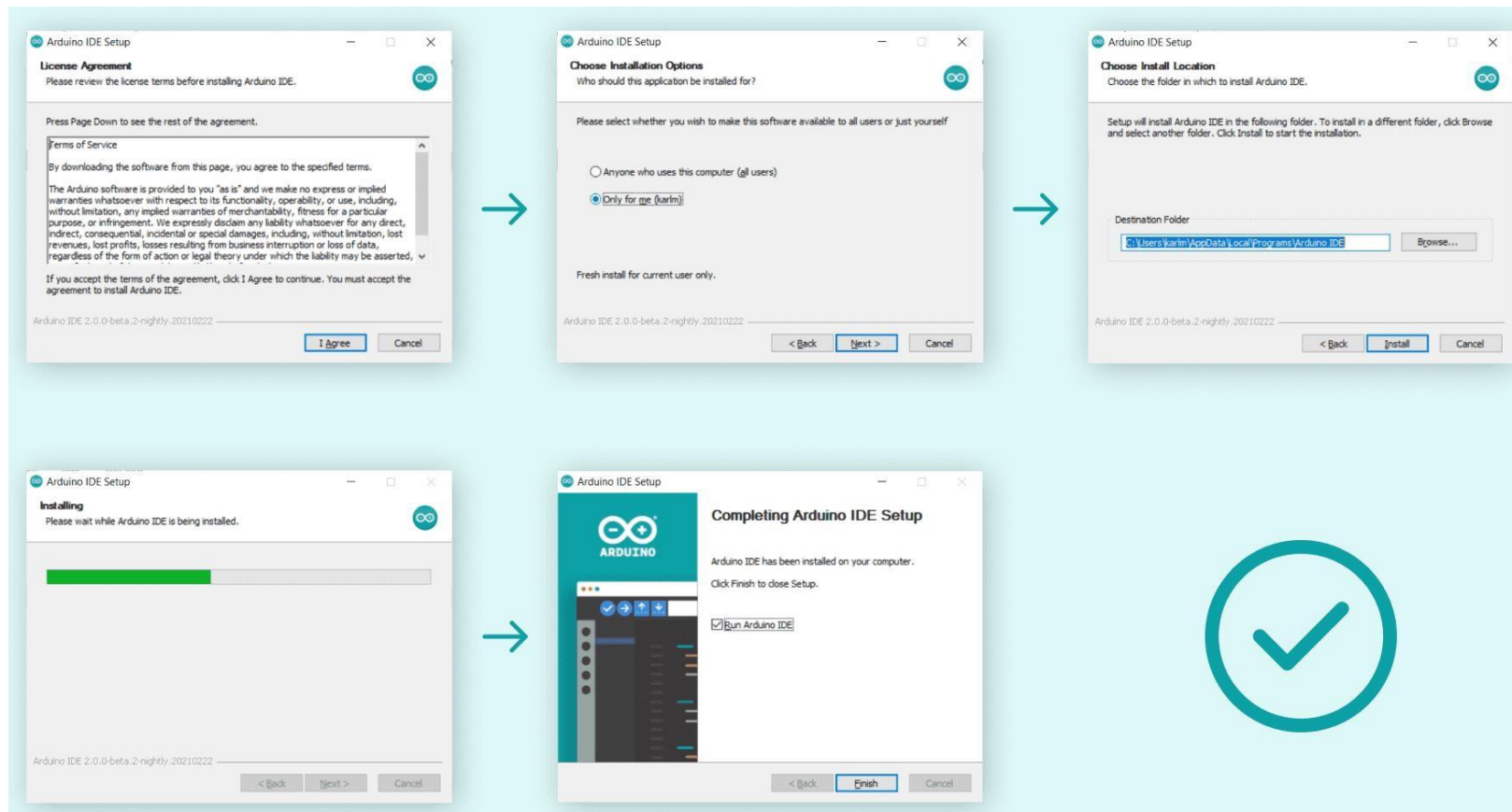
Click "JUST DOWNLOAD".

After the download is complete, you will get the installation package file with the suffix "exe"



1.3 Formal installation of Arduino IDE

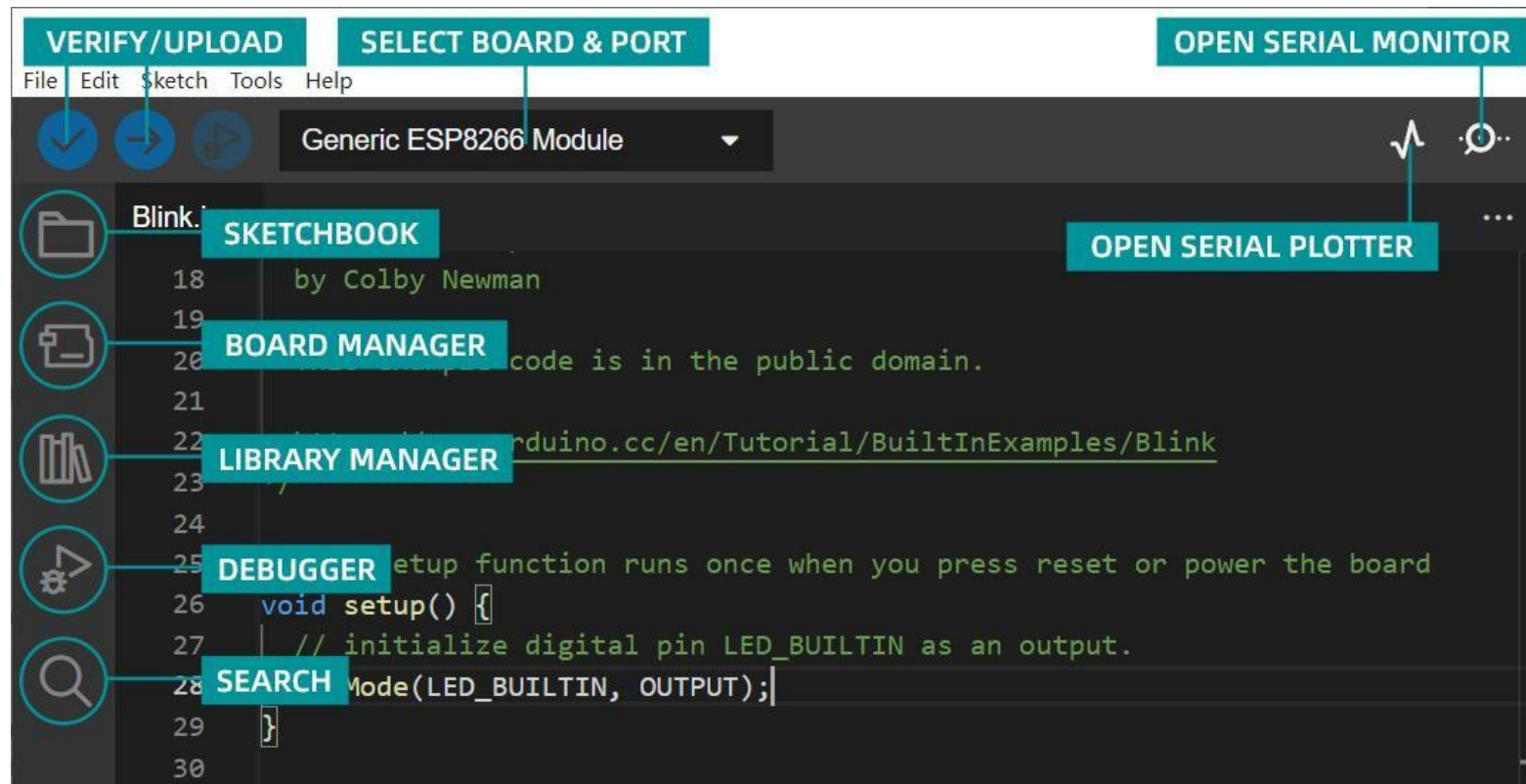
Double click to run the installer



You can press "Browse..." to select the installation path or directly enter the directory you want. Then click "Install" to install. (For Windows users, a driver installation dialog box may pop up during the installation process , when it pops up, please allow the installation)

shortcut to the Arduino IDE software will be generated on the desktop , double-click to enter the Arduino software platform environment .

After the installation is complete, open the software and you can see the software platform interface as shown below
(different versions of the interface will be different):



Each area\button function:

compile /upload - compile and upload your code to your Arduino board

Select board type and port number - The detected Arduino board and port number will be displayed here automatically

Project Sketches - Here you will find all your sketches stored locally on your computer. Also, you can sync with the Arduino cloud and get your sketches from the online environment

Board Manager - Browse Arduino and 3rd party packages that can be installed. For example, using the MKR WiFi 1010 board requires the Arduino SAMD Boards package to be installed

Library Manager - Browse thousands of Arduino libraries contributed by Arduino and its community

Debug - Test and debug programs in real time

Search - Search for a keyword in the code

Open Serial Monitor - Opens the Serial Monitor tool as a new tab in the console

A program written using the Arduino software (IDE) is called a "Sketch". These "Sketches" are written in a text editor and saved with the file extension ".ino". It is worth noting that the ".ino" file must be saved in a folder with the same name as itself. If it is not in the folder with the same name, it will be forced to automatically create a file with the same name when opening the program.

1.4 Install CH340 driver

Sometimes the computer lacks the CH340 serial port driver. Use a USB cable to connect the UNO main control board to the computer, and then search and open the "Device Manager". (If you can see CH340 under COM and LPT, you don't need to install it, just skip it)

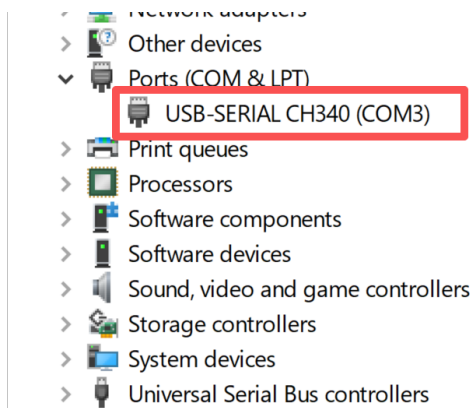


If you don't see the CH340 serial port in the above picture, you need to install the driver according to the following operations. Open the folder CH340 Driver File-Windows, double-click the exe program installation package of CH340 to

start the installation.

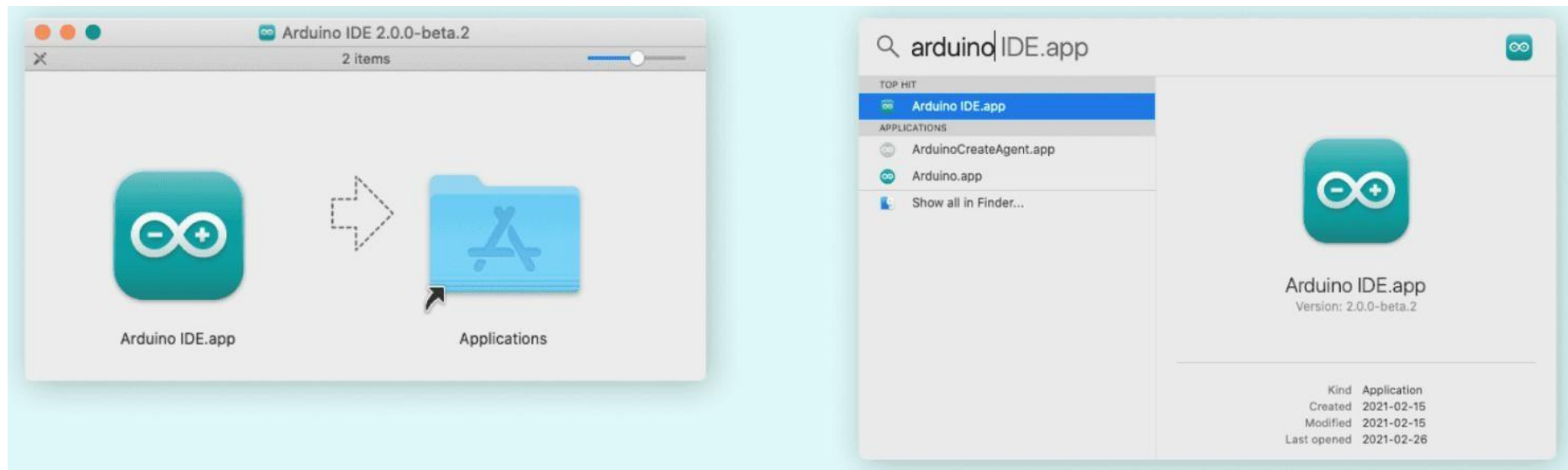
Name	Date modified	Type
2_Arduino software	2/7/2026 11:53 AM	File folder
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After the installation is complete, you can see that the driver has been displayed in the device manager (make sure the main control board is properly connected to the computer)

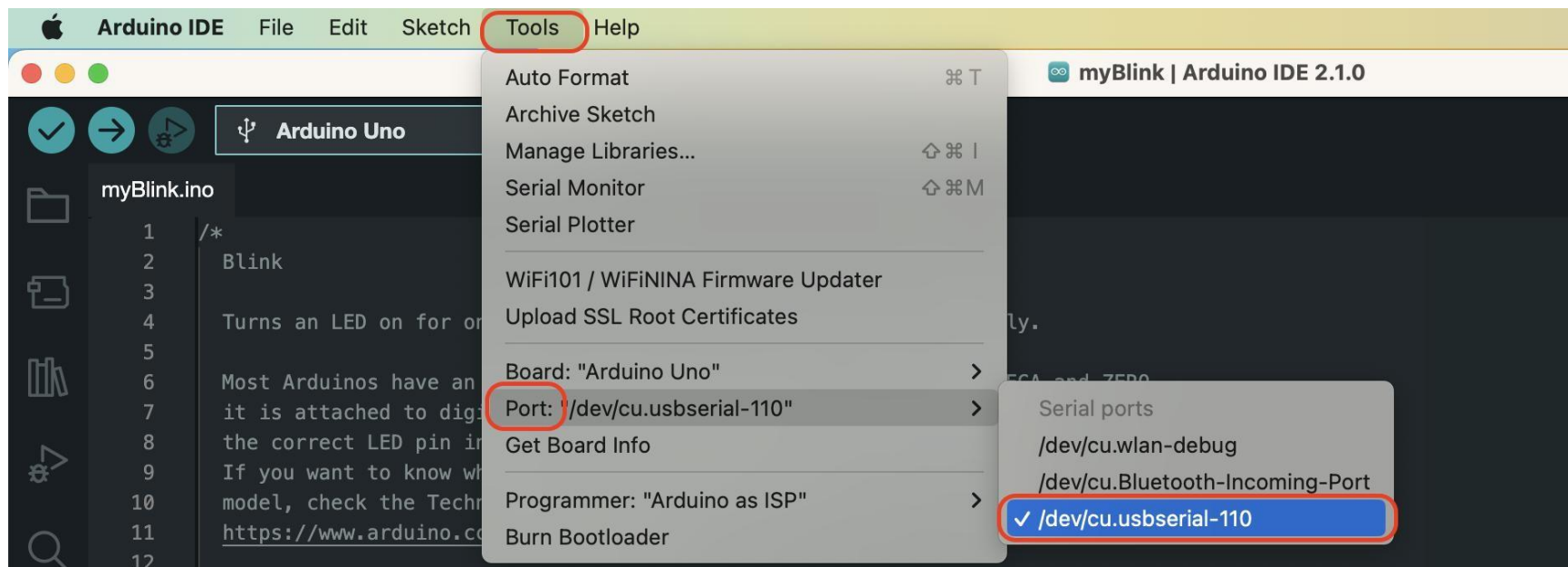


Arduino IDE on Mac OS X

Download and unzip the zip file, double-click Arduino.app to install; if there is no Java runtime library in your computer, the system will ask you to install it, after the installation is complete, you can run the Arduino IDE.



Similarly, when you connect the main control board to the computer with a USB cable, you will find that the software recognizes "USBserial" as shown in the figure below



If you do not see the USB serial port, you need to install the CH340 driver.

Open the folder CH340 Driver File-MAC, double-click to install the pkg file

Name	Date modified	Type
2_Arduino software	2/7/2026 11:53 AM	File folder
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CH340 Driver File-MAC	2/7/2026 11:53 AM	File folder
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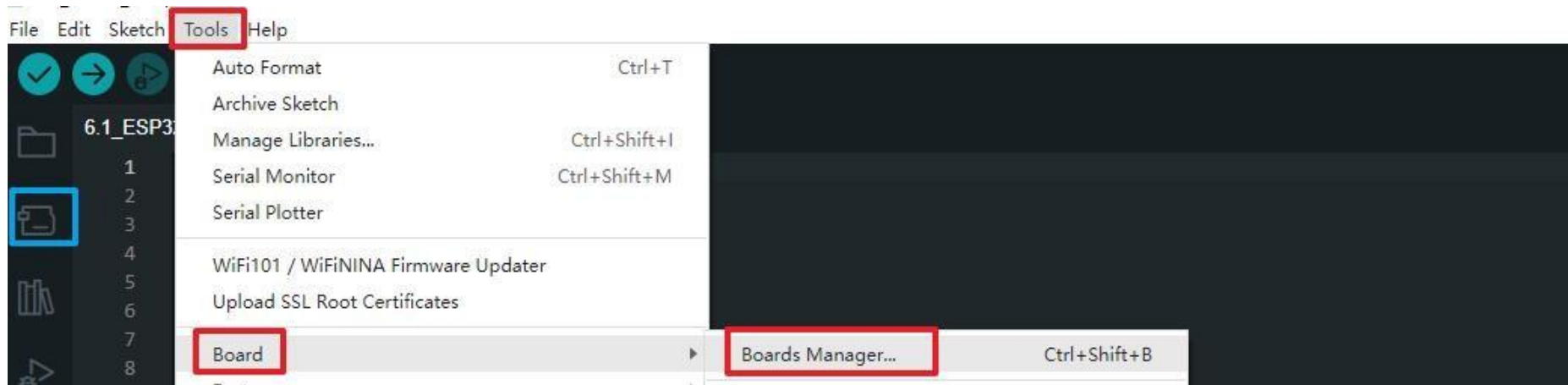
During the installation process, if the computer prompts that the installation permission is required, you need to go to the

"Security and Privacy" setting to allow the APP from any source

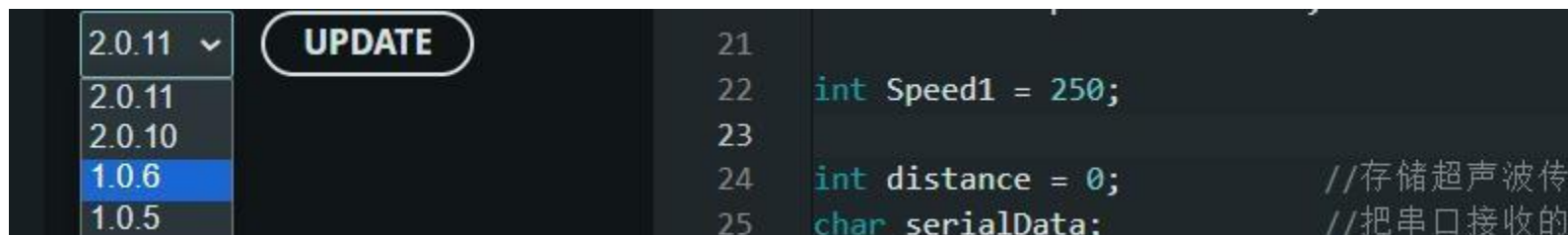
2. Add ESP32 board in Arduino IDE

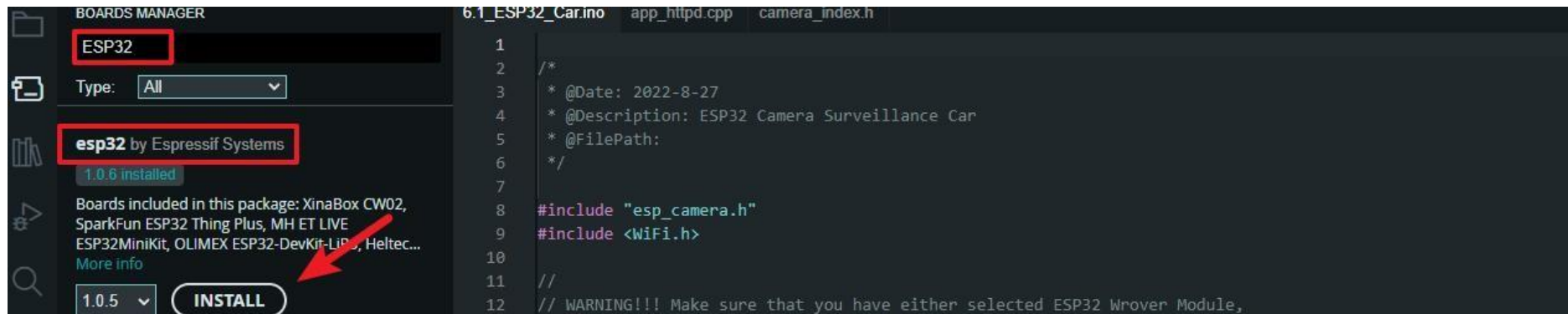
2..Search for ESP32

Open "Board Management" (the new version of IDE opens directly from the "Board Management" icon on the left)



Enter "ESP32" in the search bar, Select the latest version , and click "install" to install

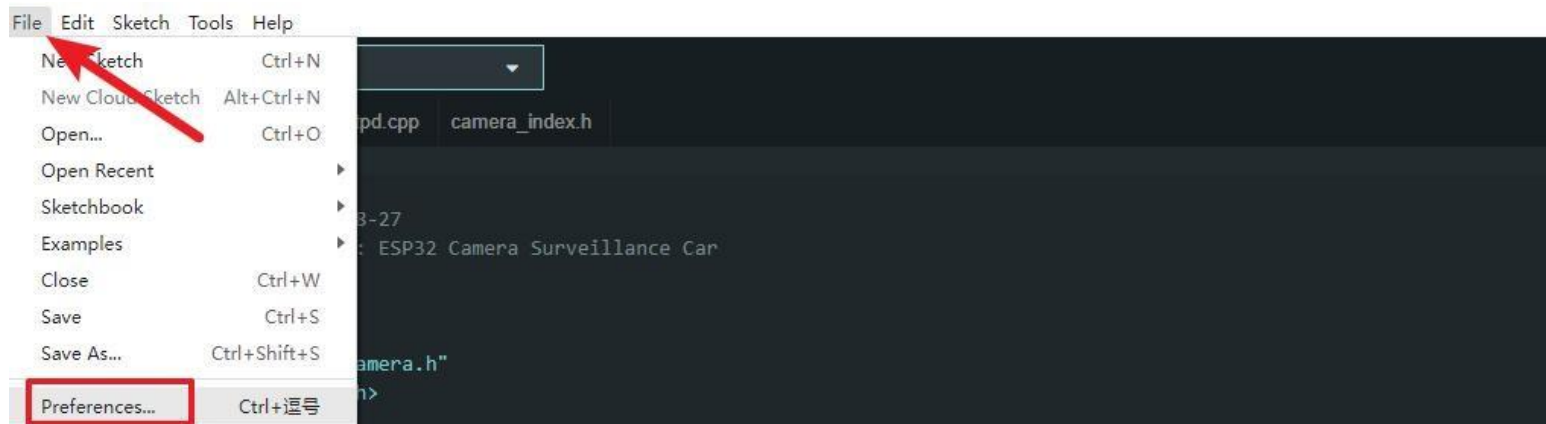




Make sure the network is open and wait for the installation to complete.

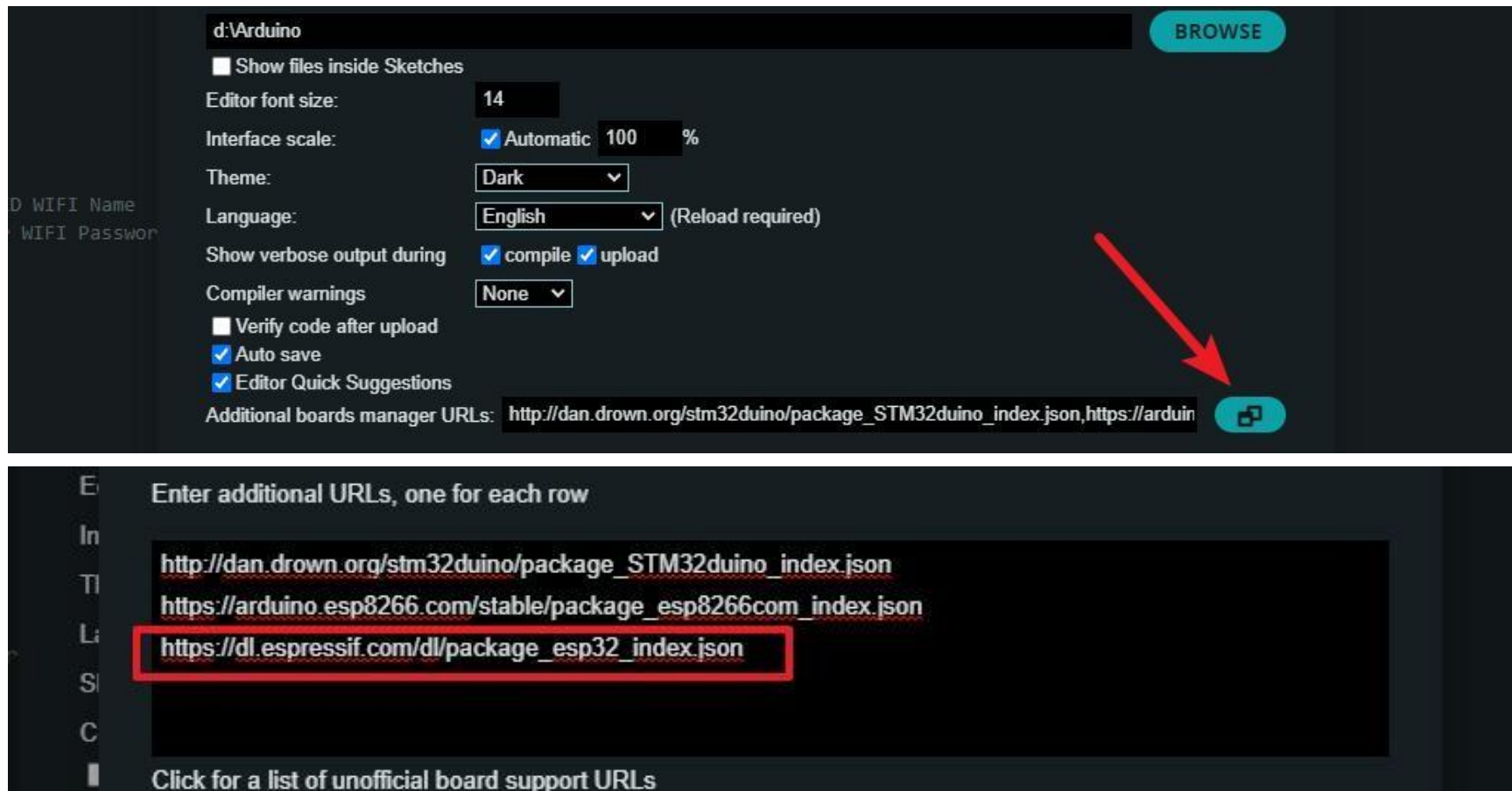
2.3. If you can't see the version list or the download is very slow?

Click "File" in the IDE and then enter "Preferences"



Copy and paste the following resource links into "Board Management URLs" and click OK to save when finished.

https://dl.espressif.com/dl/package_esp32_index.json



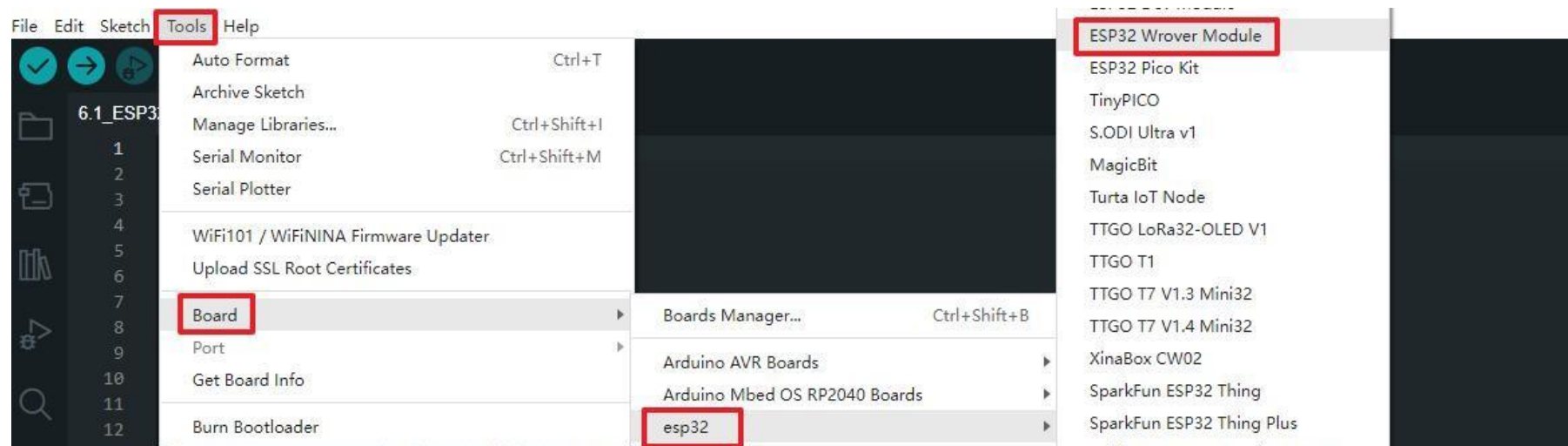
Downloading the ESP32 offline package can also install the ESP32 board:

[provide documents](#)

Intranet: [APP/esp32 2.0.9.rar](#)

2.4. Check whether the board is installed successfully

You can see "esp32" appearing under "Board"



At this point, the development environment has been created.